

7/22/23, 3:56 AM

Codio - Image Composition and Blending

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imageGen.py

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from PIL import Image

Generate two images using the DALL-E 2 API

image1_url = generate_base_image('forest')

image2_url = generate_base_image('sunset')

Download and open the generated images

image1 = download_image(image1_url, 'image1')

image2 = download_image(image2_url, 'image2')

Create a custom mask (grayscale gradient)

width, height = image1.size

mask = Image.new('L', (width, height))

for y in range(height):

for x in range(width):

mask.putpixel((x, y), x)

Composite the two images using the custom mask

result = Image.composite(image1, image2, mask)

Save the composited image to a file

result.save('composite_custom_mask.jpg')

100% (70:41)

Terminal

Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 5.19.0-1028-aws x86_64)

* Documentation: https://help.ubuntu.com

* Management: https://landscape.canonical.com

* Support: https://ubuntu.com/advantage

* Canonical Livepatch is available for installation.

- Reduce system reboots and improve kernel security. Activate at: https://ubuntu.com/livepatch

*

* Welcome to the Codio Terminal!

*

Guide

Collapse

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Image Compositing with Masks

You can also create custom masks to apply more advanced compositing techniques or control the visibility of image layers in a more precise manner. Custom masks can be created using the `Image.new()` function with the `'L'` mode, which represents a grayscale image.

Here's an example of creating a custom mask and using it to composite two images. Please note, if you are going to run it multiple time the generation and the download need to be commented out so you can keep your base image consistent.

from PIL import Image

Generate two images using the DALL-E 2 API

image1_url = generate_base_image('forest')

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result = Image.composite(image1, image2, mask)

Save the composited image to a file

result.save('composite_custom_mask.jpg')

TRY IT!

Terminated by timeout

Click here to refresh your composite_custom_mask image

Click here to refresh your 1st image

Click here to refresh your 2nd image

In this example, the custom mask is a grayscale gradient from left (black) to right (white). When using `Image.composite()`, the black areas of the mask will take the corresponding pixels from `image1`, while the white areas will take the pixels from `image2`. Gray areas of the mask will result in a blend of the two images.

Advanced Masking Techniques

You can create more complex custom masks using a variety of techniques, such as drawing shapes, adding text, or even using another image as a mask. By combining these techniques, you can create intricate and visually stunning image compositions.

When using a custom mask for image compositing, the black areas of the mask will take the corresponding pixels from `image1`, while the white areas will take the pixels from `image2`.

☒ True

☐ False

True. When using a custom mask for image compositing, the black areas of the mask will indeed take the corresponding pixels from `image1`, while the white areas will take the pixels from `image2`. Gray areas of the mask will result in a blend of the two images, allowing for smooth transitions and visually appealing compositions.

Check It!

Next

https://codio.com/sandipan-dey/image-composition-and-blending/tree/imageGen.py1/1